



Assignment 4: JavaScript Libraries for Web Mapping Applications (Mapbox)

OBJECTIVES:

This assignment aims to further enhance students' web mapping and development skills. The goal is to create a **cartographically customized thematic basemap** and develop a **simple web mapping application** using [Mapbox](#)'s modern mapping technology and JavaScript libraries. Additionally, learners will refine problem-solving skills by addressing web map constraints, such as limited token credits.

LEARNING OUTCOMES:

Assignment 4 will meet a component of learning outcomes 3 and 6 of the course. *To successfully demonstrate and fulfill the Learning Outcomes, this evaluation must meet or exceed a passing grade of 60%. Please note that resubmissions for re-evaluation will not be accepted.*

COLLABORATION FRIENDLY:

Your classmates are a valuable resource as you expand your knowledge of JavaScript libraries for web mapping applications and work toward completing this assignment. You may choose to work independently or collaborate with others. *However, each student must submit their own work by the submission deadline.*

DELIVERABLE(S):

Upon completion, **zip/compress** the assignment folder and upload it to the Assignment 4 submission folder on Brightspace.

EVALUATION:

Assignment 4 will be evaluated based on the **Assignment 4 Rubric** (attached). *Please review the rubric before submitting your final deliverable(s).*

SUBMISSION DEADLINE:

THURSDAY, FEBRUARY 20TH (4:00 PM)



TO DO

1. **Familiarize yourself with Mapbox** by reviewing the module's lecture notes and resources on Brightspace.
2. **Create an assignment folder named A4_Mapbox_YourInitials.** Maintain a well-organized folder/file structure throughout the assignment, utilizing appropriate subfolders for any input, output, and code files.
3. **Gather design inspiration** by searching online for web maps, cartographic styles, graphic designs, posters (etc.) that catch your interest. Save your selected inspiration samples in a new **inspiration subfolder**. Consider exploring the following sources for inspiration:
 - NSCC online library resources
 - Cartographer portfolios or blogs (Mike Hall, John Nelson, Daniel Huffman, Sarah Bell, etc.)
 - Social media platforms (LinkedIn, Instagram, etc.)
 - Stock photography (Unsplash, Adobe Stock, Freepik, or other open-source imagery)
4. **Select and prepare a GOOD geospatial dataset (~50-200 points)** that justifies the creation of a web map application. The dataset can originate as a shapefile, CSV, or GeoJSON, *but the final input uploaded to Mapbox must be converted to GeoJSON.*

Data Preparation Guidelines:

- ✓ **Prioritize attribute-rich datasets** to enhance pop-ups and user interaction.
- ✓ **Clean and prepare** the data to ensure data integrity, addressing gaps or missing information.
- ✓ **Project the data to WGS84 (EPSG:4326)** before exporting it to GeoJSON.
- ✓ **Remove the CRS reference** from the GeoJSON code, if present.
- ✓ **Validate the GeoJSON file** in a web-based tool like geojson.io to ensure accurate orientation and location.

Save both the original dataset and the cleaned, prepared .geojson file in your assignment folder.

5. **Develop a Web Map Application** using [Mapbox Studio](#), the [Mapbox GL JS API Reference Library](#), [Mapbox GL JS Plugins & Frameworks](#), and relevant guides and tutorials, all in combination with an IDE like VSC. Begin by cartographically designing the web map and data with hierarchy and balance, ensuring it aligns with the chosen inspirational sample(s). Then structure the map into a web application that spans the entire browser viewport.

The web map application must properly utilize an *index.html* (HTML structure), *style.css* (external stylesheet), and *script.js* (JavaScript functionality). Incorporate the best web development practices you've learned to date.

Ensure that all **Web Map Application Requirements** outlined on the next page are fully integrated.

6. **Include proper code documentation** in *appropriate locations* within the code files, including the following:
 - ✓ **Header comments** with course name, purpose, author, and date.
 - ✓ **Section comments** to distinguish major structural components.
 - ✓ **Additional comments** for user guidance where necessary.
 - ✓ **Attribution** for any externally sourced marker icon(s).
 - ✓ **Student disclaimer statement.**
 - ✓ **Copyright NSCC COGS YEAR**
 - ✓ **Citations in APA 7th edition** for data source(s) and supplementary references used.
7. **Ensure proper Mapbox attribution** is followed, based on Mapbox's [attribution documentation](#) guidelines.
8. **Final review and submission!** Proofread all displayed content, debug for error-free codes, and remove unused folders and unnecessary intermediate files.

WEB MAP APPLICATION REQUIREMENTS

In addition to the standard HTML webpage requirements learned to date, ensure the following **additional web map application requirements** are met:

- **SEO & Readability Enhancements:**
 - ✓ Include relevant **meta keywords** for improved searchability.
 - ✓ Add a relevant HTML document **title** (i.e. browser tab title).
 - ✓ Use **at least one (1)** web-friendly font (eg. Google Font).
- **Cartographic Design:**
 - ✓ Cartographically customize the **base vector tile layer** design to reflect the chosen inspiration sample(s).
 - ✓ Ensure **dynamic behaviour** of the web map (eg. generalization at smaller scales/zoom extents).
 - ✓ Style the data layer with **open-source, legible SVG marker icons** relevant to the dataset theme.
- **Enhanced User Interactivity:**
 - ✓ Use **Mapbox GL JS libraries and plugins** to improve the user's interaction with the web map application and overall user experience.
 - ✓ Implement **functioning pop-ups**. *Hint: Refer to [this tutorial](#)!* [Popup]
 - ✓ Integrate **at least five (5)** additional API functions/plugins that are suitable for the web map's purpose.

These can include, but are not limited to:

- Scale bar [[ScaleControl](#)]
- Static or dynamic legend [[mapbox-gl-legend](#) or [mapboxgl-legend](#)]
- Navigation controls [[NavigationControl](#) or [Mapbox-gl-controls](#)]
- Box zoom controls [[BoxZoomHandler](#)]
- Geolocate Search Bar [[GeolocateControl](#) or [MapboxGeocoder](#)]
- Mini-map overview [[mapboxgl-minimap](#) or [mapbox-gl-globe-minimap](#)]
- Swipe comparison tool [[mapbox-gl-compare](#)]
- Full-screen mode [[FullscreenControl](#)]

A4: JavaScript Libraries for Web Mapping Applications (Mapbox)

Course: CRTY1033/3915/Web Mapping(B)/Ejdrygiewicz,Jill

Criteria	Exceptional	Satisfactory	Unsatisfactory	Criterion Score
Organizational Competence	5 points Folder is well-structured with appropriate subfolders for input, output, and code. Unnecessary files are removed. Strong project organization is evident.	3 points Folder is organized but could be improved with minor refinements. Some unnecessary folders or files may be present. Could be improved with minor revisions/reconsiderations.	0 points Folder is disorganized or missing key elements. Best practices are not followed. Attention is required. Significantly incomplete or missing.	/ 5
Dataset Quality and Preparation	10 points Dataset is highly relevant, properly formatted (WGS 1984, GeoJSON), and well-prepared. Data integrity is maintained.	6 points Dataset is relevant but may have minor formatting or integrity issues. Cleaning and preparation are sufficient but could be improved with minor revisions/reconsiderations.	0 points Dataset is marginally relevant or has major formatting issues. Cleaning and preparation are inadequate. Attention is required. Significantly incomplete or missing.	/ 10
Data Layer Style	10 points Data layer style is highly customized, effectively complements the dataset, and uses appropriate, legible marker icons.	6 points Data layer style is customized but may lack full alignment with the dataset theme. Some icons may be unclear. Could be improved with minor revisions/reconsiderations.	0 points Minimal or no customization. Marker icons are illegible, poorly aligned with the theme, or poorly integrated. Attention is required. Significantly incomplete or missing.	/ 10
Cartographically Customized Base Vector Tile Layer	15 points Mapbox Studio customization aligns well with inspiration sources, ensuring strong visual balance and readability at different scales. Dynamic behaviour is evident at different scales or zoom extents.	12 points Base vector tile layer customization is evident but may lack full alignment with inspiration sources or dynamic behaviour. Some readability issues are present. Could be improved with minor revisions/reconsiderations.	0 points Minimal or poor customization. Weak resemblance to inspiration sources. Poor readability and usability. Attention is required. Significantly incomplete or missing.	/ 15
User Interactivity and Map Experience	15 points Mapbox GL JS libraries are effectively utilized. Pop-ups work well, and at least 5 additional API functions/plugins are seamlessly integrated. No major bugs.	9 points Web map functions with some minor interaction issues. One or more API functions may be missing or improperly implemented. Could be improved with minor revisions/reconsiderations.	0 points Minimal or ineffective interactivity. Pop-ups may not function, and API functions are missing. Attention is required. Significantly incomplete or missing.	/ 15
Code Functionality, Readability, and Quality	15 points Code is error-free, well-structured, and follows best practices. Comments are clear and comprehensive.	9 points Code is mostly functional with some minor errors. Readability could be improved, and comments may be incomplete. Could be improved with minor revisions/reconsiderations.	0 points Code has significant errors, lacks structure, and is difficult to follow. Comments are sparse or missing. Attention is required. Significantly incomplete or missing.	/ 15
Attributions and Citations	10 points Proper Mapbox, marker icon, and dataset attributions are included in APA 7th edition format.	6 points Some attributions are included but may not fully meet APA 7th edition format. Could be improved with minor revisions/reconsiderations.	0 points Attributions are missing or incorrectly formatted. Attention is required. Significantly incomplete or missing.	/ 10

Total	/ 80
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Overall Score

Exceptional 72 points minimum	Good Effort 64 points minimum	Fair Effort 48 points minimum	Unsatisfactory 0 points minimum
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